

Indoor air quality and energy performance of air-conditioned office buildings in Singapore

Abstract An integrated indoor air quality (IAQ)-energy audit methodology has been developed in this study in Singapore, which provides a rigorous and systematic method of obtaining the status-quo assessment of an 'IAQ signature' in a building. The methodology entails a multi-disciplinary model in obtaining measured data pertaining to different dimensions within the built environment such as the physical, chemical, biological, ventilation, and occupant response characteristics. This paper describes the audit methodology and presents the findings from five air-conditioned office buildings in Singapore. The research has also led to the development of an indoor pollutant standard index (IPSI), which is discussed in this paper. Other performance indicators such as, the ventilation index and the energy index as well as the building symptom index (BSI) are also presented and discussed in the context of an integrated approach to IAQ and energy. Several correlation attempts were made on the various symptoms, indoor air acceptability, thermal comfort, BSI and IPSI, and while BSI values are found to correlate among them as well as with IAQ and THERMAL COMFORT acceptability, no such correlation was observed between BSI and IPSI. This would suggest that the occupants' perception of symptoms experienced as well as environmental acceptability is quite distinct from IAQ acceptability determined from empirical measurements of indoor pollutants, which reinforces the complex nature of IAQ issues.

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Key words Air-conditioned office building, Building symptom index; Field measurements; Indoor air quality audit; Indoor pollutant standard index; Questionnaire analysis; Tropical building.

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Practical Implications

This paper reports comprehensive indoor air quality (IAQ) and energy studies in a hot and humid climate and presents data from both objective and subjective measurements. Performance indices, such as ventilation index, energy index and the newly introduced indoor pollutant standard index (IPSI) are envisaged as practical tools for a quick assessment in exploring the integrated issues of IAQ and energy in buildings. Based on a detailed analysis of all the available data, it is found that building symptom index (BSI) values are found to correlate among them as well as with IAQ and THERMAL COMFORT acceptability, but no significant correlation exists between BSI and IPSI. This implies that the occupants' perception of symptoms experienced as well as environmental acceptability is quite distinct from IAQ acceptability determined from empirical measurements of indoor pollutants, which reinforces the complex nature of IAQ issues.

Introduction

The advent of the energy crisis in the early 1970s led to several concerns related to poor indoor air quality (IAQ) in buildings. Problems associated with poor IAQ and sick building syndrome (SBS) in the Western countries reached a peak in the early 1980s and, often, could be associated with inadequate ventilation. A European project was started at the end of 1992, in which current methods and trained panel were used to investigate office buildings all over Europe. The main aim of this project was to develop assessment

procedures and guidance on ventilation and source control, which help to optimize energy use in office buildings while assuring IAQ. The results, conclusions and recommendations of the audit performed in 56 buildings are summarized in a final report, the key conclusion being that occupants are a less dominant pollution source in comparison with building materials and components in the ventilation system (Bluyssen et al., 1995, 1996). A similar study in the United States, entitled Building Assessment Survey and Evaluation (BASE) programme, was aimed at collecting baseline data characterising public and

commercial office buildings (EPA, 1994a; Womble et al., 1994). One of the core parameters measured was the volatile organic compounds (VOCs), which was measured in indoor and outdoor air at 41 randomly selected public and private office buildings in 10 climatic regions in the continental United States (Hadwen et al., 1997). These buildings were randomly selected without regard to IAQ concerns but excluded any buildings with high profile IAQ issues.

Studies involving the investigation of the prevalence of occupant symptoms, such as the Danish Town Hall study and the Swedish Office Illness Project have reported that a significant proportion of occupants in non-industrial buildings may be exposed to indoor environmental conditions that can cause discomfort, lower performance levels and even adverse health effects (Skov et al., 1987; Stenberg et al., 1994). Environmental studies in moldy office buildings including typical concentrations of fungi and total volatile organic compounds (TVOC) have been attempted in the past (Morey, 1990; Morey et al., 1986; Morey and Jenkins, 1989). Numerous studies, based on questionnaire survey, have been carried out and the primary objective of all these studies has been the evolution of occupants' perception and their associated symptoms to types of indoor climate defects and deficiencies (Burge et al., 1990; Norlen and Andersson, 1993; Rowe, 1993; Wilson et al., 1987).

The issue of IAQ has received significant attention in Singapore over the past few years and considerable resources have been channelled in recent times to

better understand these issues in the local context (Chan, 1999; Chan et al., 1995; Ooi et al., 1995; Sekhar et al., 1997; Tham et al., 1997). Combined findings from these studies together with inputs from relevant professional and governmental organizations resulted in the release of local IAQ guidelines for office premises by the Ministry of the Environment in Singapore (ENV, 1996). Several studies were undertaken over the past 10 years, the scope of which ranged from measurement of indoor pollutants to the procurement of occupant responses. IAQ questionnaire analysis has been mainly concentrated on the use of multiple regression analysis (MRA) in modeling the occupants' IAQ perceptions. Many applied an integrated approach to evaluate the IAQ status of their respective case studies through the analysis of subjective and objective data in correlation. More recently, energy dimension has been included as an integral component, which has resulted in the development of an integrated IAQ-energy audit methodology. This paper presents the findings from a comprehensive study of five air-conditioned buildings in Singapore and includes measurements of exposures and technical characteristics as well as occupants' perception and symptoms.

Theoretical framework

A brief description of the theoretical framework adopted in the analysis of the different dimensions of the IAQ-energy audit methodology (Figure 1) is now discussed.

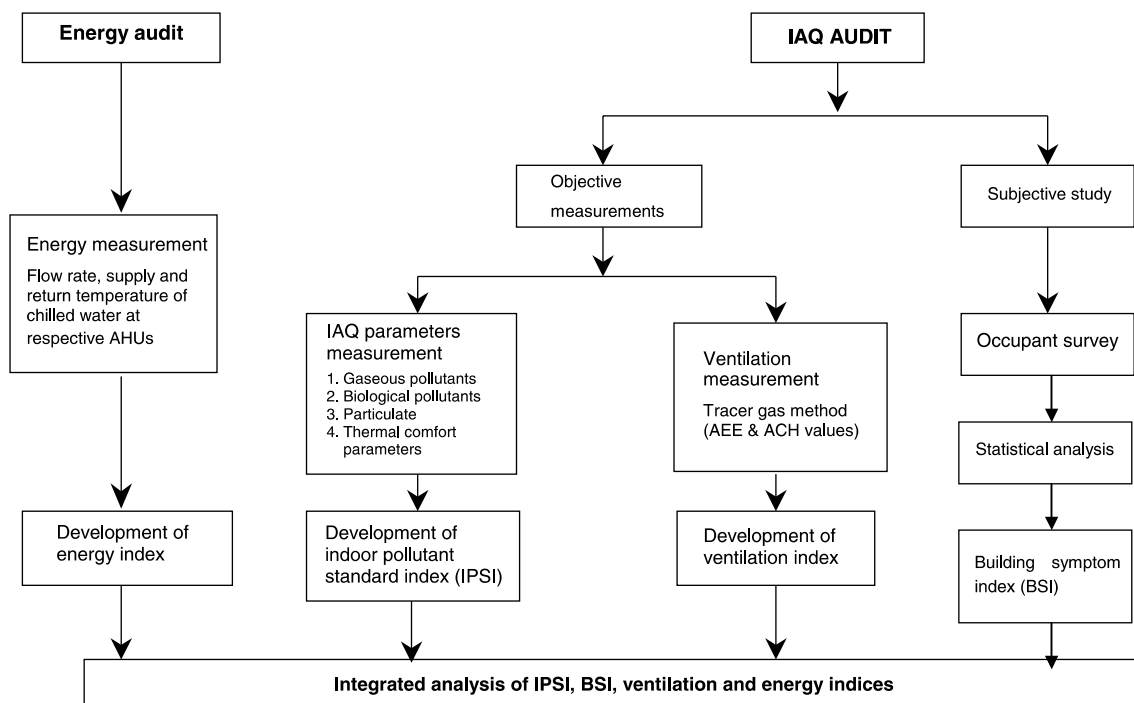


Fig. 1 Integrated IAQ – energy audit methodology

Indoor pollutant standard index (IPSI)

The measured concentration levels of the physical, chemical and biological pollutants are compared with relevant local and international standards/guidelines to establish the IAQ status of the building. In particular, continuous chemical monitoring data of carbon dioxide (CO₂), carbon monoxide (CO), formaldehyde (HCHO) and total volatile organic compounds (TVOCs) allows the profiling of the prevailing concentration levels over a weekly period, resulting in an 'IAQ signature'. The recommended threshold limit values (TLVs) specified in the local IAQ guidelines (ENV, 1996), are indicated as Breakpoint 1 values in Table 1. The local recommended values are fundamentally similar to those found in European and American standards (ASHRAE, 1999).

In order to express the quality of indoor air in different premises, we have proposed the IPSI, which recognizes that the characteristics of indoor air pollutants are significantly different from each other and their effect on human health and comfort are also different from one another (Tin, 1998; Sekhar et al., 1999). The basic concept of the IPSI is the same as that adopted for the pollutant standard index (PSI), developed by the Environmental Protection Agency, United States of America (EPA, 1994a,b). PSI is expressed as air quality of outdoor environment and many countries use it as an ambient air quality standard.

The IPSI is fundamentally based on the principle of PSI. The index is concerned with eight parameters, including HCHO, TVOCs, CO, CO₂, particulate matter, total bacteria, fungi and thermal comfort. In order to express a single parameter for thermal comfort status in indoor environment, predicted percentage dissatisfied (PPD) values are calculated based on Fanger's thermal comfort model using measured thermal comfort parameters such as dry bulb temperature, mean radiant temperature, relative humidity and air velocity (Fanger, 1970).

The IPSI is based on measured values of IAQ parameters, which are transformed onto a scale ranging from 0 to 10. For each pollutant, a sub-index is calculated from a segmented linear function that transforms the indoor concentration level onto a scale extending from 0 to 10, with two corresponding to the ENV's IAQ guideline value except thermal comfort and 10 corresponding to significant harm level of human health and discomfort. The major breakpoint concentrations used in defining the significant harm level of human health and discomfort and acceptable level of each parameter are presented in Table 1. The limit of acceptable IAQ on this scale is 2, as this limit corresponds to the acceptable threshold value established by the Ministry of the Environment, Singapore (ENV, 1996). For example, the acceptable threshold concentration of carbon monoxide in indoor air is set at 9 ppm and the detection of this level in a particular indoor air sample will be translated to IPSI sub-index value of 2. The maximum sub-index value of all the eight indoor pollutants in a particular set of data will be reported as the IPSI value.

It is pertinent to note that the present development of IPSI is restricted to two linear segments of 0–2 and 2–10. It would be ideal to refine the index scale to represent a better resolution on the scale of 0–10, particularly beyond 2. This is possible if there is consensus among the scientific community on the identification of defined points on the scale based on established epidemiological effects and/or discomfort. Unfortunately, the detailed information of such dose–response studies of indoor pollutants and scientific consensus has not been reported in the literature. It is also to be noted that while the IPSI value is meant to be an aggregate measure of IAQ acceptability it is based on a single indoor pollutant with the highest scaled value. While such simplification can underestimate the cumulative effect of several indoor pollutants, the IPSI provides a quick and easy gauge of the acceptability of IAQ based on measured concentration levels of indoor pollutants.

Table 1 Summary of indoor air quality (IAQ) parameters in all the buildings and breakpoint of IAQ parameters for IPSI

IAQ parameters	Break point 1	Break point 2	Building				
			A	B	C	D	E
Carbon dioxide (p.p.m.)	1000	35,000	511–716 (608)	429–568 (506)	442–682 (569)	595–1400 (797)	475–1400 (716)
Carbon monoxide (p.p.m.)	9	50	0.47–1.23 (0.8)	0.2–1.6 (0.7)	1.17–2.64 (1.6)	1.2–1.8 (1.4)	0.4–2.4 (1.5)
Formaldehyde (p.p.m.)	0.1	30	0.004–0.086 (0.02)	0.0046–0.02 (0.0088)	0.008–0.016 (0.011)	0.018–0.15 (0.084)	0.011–0.132 (0.057)
Total volatile organic compounds (p.p.m.)	3	6	0.78–1.31 (0.97)	0.1–0.8 (0.35)	1.52–2.26 (1.85)	1.68–2.75 (2)	1.6–2.5 (2.25)
Total bacteria (O/L) (CFU/m ³)	500	3000	18–500 (421)	9–396 (147)	150–89 (430)	106–1214 (403)	114–528 (318)
Total bacteria (A/S) (CFU/m ³)	500	3000	26–175 (118)	9–387 (169)	26–1021 (269)	106–660 (304)	35–572 (196)
Total yeast and mould (O/L) (CFU/m ³)	500	3000	25–420 (147)	9–528 (135)	44–290 (113)	26–660 (174)	53–686 (244)
Total yeast and mould (A/S) (CFU/m ³)	500	3000	26–569 (129)	9–528 (166)	9–343 (64)	35–352 (131)	18–625 (243)
Particulate matter (µg/m ³)	150	600	2–98 (31)	23–117 (106)	9–104 (29)	10–49 (25)	9–44 (20)
Thermal comfort [PPD (%)]	20	100	8–34 (19)	7–23 (17)	7–17 (13)	6–18 (14)	12–25 (20)

Mean values are given in parentheses.

Breakpoint 1: acceptable level (scale value = 2).

Breakpoint 2: significant harm level (scale value = 10).

Ventilation characteristics and ventilation index

Tracer gas techniques are employed to determine the ventilation characteristics such as the air change per hour (ACH), Air exchange effectiveness (AEE) parameters and the global air exchange efficiency (Fisk et al., 1993; Sekhar and Tham, 1995). These are defined as follows:

$$AEE_{GLOBAL} = AEE_G = t_N / \langle t \rangle \quad (1)$$

$$AEE_{OCCUPANT LEVEL} = AEE_{OL} = t_N / \langle t_{OL} \rangle \quad (2)$$

$$AEE_{LOCAL} = AEE_L = t_N / t_{OL} \quad (3)$$

$$n_{GLOBAL AIR EXCHANGE} = n_{ACG} = t_N / (2 \times \langle t \rangle) \quad (4)$$

where t_N = the indoor air volume divided by the outside air flow rate and is the reciprocal of ACH, t_{OL} = local age of air measured at the occupant level, $\langle t \rangle$ = room average age of air within the entire cluster of zones and determined from measurements of tracer gas in the main return or exhaust air streams, $\langle t_{OL} \rangle$ = average of the measured local ages of air at occupant level.

The ventilation index is based on the actual amount of fresh air provided, calculated from measured air change rates (ACH). The index in terms of litres per second per square metre (lps/m^2) for each location is normalized on the basis of air-conditioned floor area.

Energy index

The IAQ-energy audit methodology is based on a system-zone concept, which implies that all the indoor sampling points for a measurement set-up belong to a zone or set of zones that are served by the same air-conditioning and ventilation system. In the event of a set of zones that constitute a control volume of space being served by more than one air handling unit (AHU), then all the relevant AHUs serving the set of zones are included in the study as part of the same set-up. Such a 'system-zone' concept was employed to ensure that the prevailing IAQ levels in the occupied zones could be attributed to the characteristics of the air-conditioning and the air distribution system. This was also envisaged to facilitate a study of the energy consumption at the micro level associated with a certain level of IAQ.

The energy index is developed by computing the cooling energy consumption of the respective AHU and is obtained as follows:

$$Q = (mc_p \Delta t) \quad (5)$$

$$EI = (Q \times n) / \text{Total floor area} \quad (6)$$

where Q = cooling capacity in kW, EI = energy index in $kWh/m^2/\text{month}$, m = mass flow rate of chilled water, Δt = chilled water temperature difference

through the cooling coil, c_p = specific heat of water, and n = number of hours of operation of the air-conditioning system.

The following assumptions have been made to represent the energy index,

- Calculated cooling capacity (Q in kW) is an average measure of capacity and the energy consumption (kWh) is an average rate of energy consumption.
- Expected hours of air-conditioning system operation per month is 195 assuming 8 h of operation on week days and 5 h of operation on Saturdays.

Building symptom index (BSI)

In the area of perceived IAQ and comfort, the use of a validated questionnaire is very important and the questionnaire used in this study was modeled after the one used in the 'European Audit Project to Optimize IAQ and Energy Consumption in Office Buildings' (Bluyssen et al., 1995; Roulet et al., 1994). Occupants are surveyed on a range of questions, which includes their work details, health status during the 'past month' and at present and how they perceived their environmental conditions during the 'past month' and at present. Comments were also solicited on whether they had individual control of their office environment. The definition of BSI, employed in this study is similar to that in the EC-Audit and BASE 95 studies. It is an indicator of the well-being and health condition of the occupants as deduced from a questionnaire survey. It is the average number of sickness symptoms declared per person, out of a selected list. In deriving the BSI, only the symptoms that disappear after leaving the building are considered, in conformance to the notion of SBS terminology (Roulet, 1994).

The building-related symptoms, namely dry eyes, block nose, dry throat, dry skin, headache, lethargy, watery eyes, runny nose, chest-tightness symptoms, flu-like symptoms, rash and other symptoms are classified in accordance to the EC-audit procedures. The full list of symptoms consists of all the symptoms while the short list of symptoms are the first six symptoms listed. They are classified as the most prevalent six building-related symptoms experienced by most occupants.

The BSI was further sub-categorized for symptoms experienced, based within a defined time frame. Normally, the time frame is differentiated as symptoms experienced for the 'past month' (EC-Audit) or 4 weeks ago (BASE 95) and similarly for 'here and now' perception (EC-Audit) or within this week (BASE 95). The method for deriving BSI for the 'here and now' period, however, is based on a severity scale given in the relevant section of the questionnaire.

For the present study, the EC-Audit method of categorization has been adopted, namely:

- **BSIf_f** (f) based on the frequency scale, (f) with a full list of symptoms;
- **BSIf_s** (f) based on the frequency scale, (s) with a short list of symptoms;
- **BSIs_f** (s) based on the severity scale, (f) with a full list of symptoms;
- **BSIs_s** (s) based on the severity scale, (s) with a short list of symptoms.

The BSI values are also computed separately for male and female occupants to explore the differences in perceptions between genders, as this has been observed to be a significant consideration in earlier studies (Stenberg et al., 1994).

Methods

Integrated IAQ-energy audit methodology

The integrated IAQ-energy audit methodology is illustrated in Figure 1, depicting the generic stages and their associated investigative procedures and methods (Sekhar et al., 1995; Tham, 1994; Tham et al., 1996). The measured IAQ and ventilation parameters are transformed into IPSI and ventilation index respectively. The computed energy data is converted into an energy index, which is obtained as the total chilled water energy consumption divided by the occupied floor area. The subjective responses of the occupants are used in determining BSI, which essentially reflect the well-being of the occupants. These indices could be construed to provide a basis for a comparative evaluation of the performance of buildings in as much as achieving and sustaining a certain level of IAQ is concerned.

Measurement procedures

The procedures for obtaining the objective and subjective information pertinent to a typical case study are now discussed. After examining the building and its systems' functions, design and operation and maintenance, the number of zones for carrying out IAQ investigations is determined to provide a balance between coverage and cost. For each zone, the sampling locations are then selected based on occupant and equipment densities, layout, partitioning, furniture, activities and processes, and the ventilation and air distribution strategy.

A multigas monitor (Brüel & Kjær, Naerum, Denmark), based on the photo-acoustic infra-red detection method, is used in conjunction with a multi-point sampling system to obtain continuous real-time chemical data. Continuous real-time monitoring of CO₂, CO, HCHO and TVOCs are carried out at five

indoor points and one ambient point. A simultaneous 6-h measurement of formaldehyde at the various indoor locations was obtained employing dinitrophenylhydrazine (DNPH) cartridges and subsequently analysed using the high performance liquid chromatography (HPLC) method. Owing to the high threshold detection of the infra-red photo-acoustic instrument, the values of formaldehyde obtained by the multigas monitor were observed to be inconsistent and in some cases quite unreliable. As a result of this observation, the results of formaldehyde presented in this paper are those obtained through the HPLC method. A single-stage air impaction sampler is used to assay biological samples. Air samples for culturing of bacteria and yeast and mold are collected at the designated locations at 1.2 m above ground, air diffusers and outdoor. Trypticase soy agar (TSA) and potato dextrose agar (PDA) are the culture media used for the total bacteria and the total yeast and mold respectively. The thermal comfort level of the indoor environment is measured using an integrated indoor climate analyzer. It is a portable instrument that senses environmental conditions via five separate transducers, simultaneously. The measurements taken through the transducers include room air temperature (dry bulb), relative humidity, air velocity and radiant temperature. The concentration of the suspended particulate is measured using an environmental laser aerosol spectrometer. This is capable of gravimetric measurements ($\mu\text{g}/\text{m}^3$) of particulate, measuring particles between 0.1 and 15 μm in aerodynamic diameter, with an accuracy of $\pm 5\%$.

Ventilation study involves the measurements of the air exchange rate, quantity of fresh air, age of air, AEE and air exchange efficiency. The various ages of air, air change rate (ACH) and AEE values are determined from tracer gas studies, using a multigas monitor and a multiple doser and sampler, in which a tracer gas (sulfur hexafluoride, SF₆) is dosed into the fresh air stream and its concentration at various sampling points in the occupied zones as well as the return air path is monitored.

Energy is computed from measurements of chilled water temperatures and chilled water flow rates. In situations not equipped with built-in flowmeters, an ultrasonic flowmeter (Panametrics Inc., Waltham, MA, USA) is used to measure the chilled water flow rate in a non-destructive manner. In cases involving the use of building automation system (BAS), these parameters are directly obtained from BAS reports.

For occupant perception, their evaluation of the physical environment and physiological responses are solicited through a comprehensive survey questionnaire which is augmented from that adopted in the EC Audit study (Bluyssen et al., 1995; Roulet et al., 1994). It was envisaged that the use of a questionnaire modeled after the one used in the EC-Audit study would enable a comparison in the perception of IAQ

between the European and the Singapore studies, which is being developed. A total of 426 completed questionnaires were received, which is ideal for the extensive use of modeling techniques.

Description of case studies

A total of five buildings were studied during the course of the current research project. Building A is an institutional building in which the measurement process involved seven different set-ups. Each set-up constitutes a study that consists of a dedicated AHU serving one or more occupied zones so that the concept of system-zone is employed in all the studies. Building A employs a floor by floor AHU concept in which the various AHUs and the occupied zones that they serve are distinctly separated. A setup would also typically involve five indoor sampling locations with an ambient location for monitoring and an AHU-return air location for ventilation measurements. Thus, building A involves a total of 42 sampling locations. Buildings B, C, D, and E are all high-rise office buildings in the Central Business District area. Building B has a square floor area and employs floor by floor AHUs. Six set-ups involving 30 sampling locations were carried out. Building C has a more rectangular floor plan with two AHUs per floor. Three set-ups involving 18 sampling locations were studied. Building D is also a square building with dedicated floor by floor AHUs. Three setups involving 15 sampling locations were investigated in building D. Building E has a curved floor plan with a dedicated AHU in each floor in which three setups involving 15 sampling locations were studied. The detailed measurements of buildings A, B, C, D and E are summarized and discussed in this paper. The ventilation provision in A, B and C is by means of fresh air openings at the respective AHUs in each floor while D and E employ a centralized fresh air duct with branch ducts to the individual AHUs at each floor. All the five buildings employ chilled water-based AHU systems and the return air path is through ceiling plenum.

Results and discussion

An integrated analysis of the results of all the five buildings investigated in the research project is now presented under the following headings:

- Objective measurements
- Subjective measurements
- Performance indices

In the case of objective measurements, the results are a summary of all the different set-ups of measurements in each building. The prevailing pollutant levels are compared against recommended local IAQ guidelines (ENV, 1996; BCD, 1983; BCD, 1979; ASHRAE, 1992). The ventilation characteristics

in terms of the provision of fresh air as well as the AEE parameters for the five buildings are presented and discussed. In the case of subjective measurements, the results are presented as responses of building symptoms by the occupants for both the 'past month' and the 'here and now' periods. The subjective response analysis is also characterized for gender differences. The IPSI developed in this project is computed for all the buildings as one of the performance indices. This is discussed in conjunction with the other performance measures such as the energy index and the ventilation index, which are also computed for all the five buildings. In the case of subjective response data, BSI values are computed separately for female and male occupants for the five buildings. The various performance indices are compared across all the five buildings to provide a better insight into the relevance and usefulness of such indices in establishing a quick gauge of the overall quality of air in the built environment.

Objective measurements

A summary of the chemical and biological pollutants as well as the particulate levels and measured thermal comfort values in all the five buildings is presented in Table 1. It is seen from the range and the mean values that the measured IAQ parameters are generally within recommended TLV with some exceptions. It is also seen from the thermal comfort parameters in Table 2 that the temperatures are slightly inclined to the colder side. The ventilation characteristics, such as the ACH, fresh air provision and the AEE parameters are presented in Table 3. It is seen that the ACH ranges quite substantially amongst the five buildings investigated. It is also seen that the fresh air provision on the basis of design occupancy is generally acceptable in most buildings, except relatively low provisions in certain parts of building E. The AEE parameters are also generally indicative of a well-mixed flow pattern, which is typical of conventional designs of air distribution systems with high recirculation rates. The cooling capacities and energy consumption of these AHUs are calculated and presented in Table 4. The cooling capacity ranges between 34.5 and 213 kW and the energy consumption vary between 8 and 83.4 kWh/m².

Table 2 Summary of average values of thermal comfort parameters in all the buildings

Building	Dry bulb temperature (°C)	Radiant temperature (°C)	Humidity (%)	Air velocity (m/s)
A	23.8	24.6	60	0.18
B	22.6	23.2	64	0.13
C	22.6	23.4	59	0.15
D	21.7	22.9	57	0.17
E	23.6	24.1	51	0.14

Table 3 Summary of ventilation parameters in all the buildings

Building	Air change per hour	Fresh air (lps/person) (design occupancy ¹)	Fresh air (lps/person) (actual occupancy ²)	AEE _G	AEE _L	AEE _{OL}
A	1.00–2.60	5.7–16.9	3.7–42.3	0.96–1.29	0.96–1.11	0.97–1.02
B	0.72–1.17	4.4–7.7	4.9–11.5	1.24–1.27	0.94–1.04	0.98–1.0
C	0.67–1.34	3.6–7.9	5.5–12.4	1.18–1.32	0.58–1.13	0.74–1.11
D	0.55–0.57	3.5–3.8	NA	1.46–1.80	0.96–1.03	0.99–1.00
E	0.34–0.59	2.3–4.0	NA	1.33–1.52	0.83–1.23	0.85–0.99

¹ Based on the value of 10 m² per person.

² Refers to actual number of people in the location.

Global air exchange effectiveness (AEE_G), localized air exchange effectiveness (AEE_L) and occupant level air exchange effectiveness (AEE_{OL} = 1) and ($n_{ACG} = 50\%$) imply perfectly mixed flow pattern and could thus be considered good from a dilution perspective. Such flow patterns are normal design practices in office air-conditioning systems. Values of AEE_G, AEE_L and AEE_{OL} > 1 and $n_{ACG} > 50\%$ are even better as they indicate increased dilution. AEE significantly < 1 is indicative of short-circuiting.

Table 4 Summary of energy measurements in all the buildings

Building	CHWS (°C)	CHWR (°C)	V (m/s)	m (l/s)	Q (kW)	E (kWh/m ²)/month
A	7.0–8.5	12.5–14.5	0.95–1.60	3.17–4.91	68.0–98.0	38.8–83.4
B	6.0	16.0	1.14	3.23	133.0	20.8
C	6.3–6.7	8.6–9.4	NA	3.42–6.70	34.5–70.0	11.6–23.7
D	6.3	14.1	NA	6.49	213.0	25.8
E	9.2–10.4	17.9–22.0	0.26–0.38	1.39–2.40	53.5–117.0	8.0–20.2

CHWS, chilled water supply temperature; CHWR, chilled water return temperature; V, chilled water velocity; m, chilled water flow rate; Q, cooling capacity = mc_p (CHWR – CHWS) where $c_p = 4.2$; E, energy consumption = (cooling capacity × No. of AHU operating hours)/occupied floor area = cooling capacity × 195.

(Monday–Friday = 8 h/day; Saturday = 5 h; [(8 × 5 days/week × 52 weeks/year) + (5 × 52)]/12 = 195 h.

Subjective measurements

It is to be noted that the proportion of male to female respondents is rather uneven, with approximately 64% of the respondents being female. The mean age of the respondents for the five audited buildings is 32.74 years. The SBS analysis of the full list of 12 symptoms, for ‘past month’ and ‘here and now’, are stated as the percentage of the occupants in each building complaining of individual symptoms and these are further distinguished for male and female occupants. The results for all the five buildings showing percentage of occupants (total, female and male separately) experiencing each symptom for the ‘past month’ are presented in Figure 2. Similar results for the ‘here and now’ period are shown in Figure 3. It is seen quite clearly that the female occupants register significantly higher building-related symptoms in comparison with their male counterparts in both the ‘past month’ and the ‘here and now’ periods. Building-related symptoms by female occupants are higher than male occupants typically by a factor of 1.5–2, and in one case (dryskin for ‘here and now’ period) as high as 4.2. It is also observed that the six predominant symptoms for ‘past month’ and ‘here and now’ periods are the same and includes dry eye, block nose, dry throat, dry skin, headache and lethargy, although the percentage and ranking differ.

The occupants’ perception of the indoor air varies amongst the five buildings, and the mean indoor air

acceptability values based on the 10-point scale are presented in Figure 4. The indoor air acceptability based on percentage dissatisfied is presented in Figure 5. Occupants’ perception of the thermal comfort level in the five buildings is presented in Figure 6 and it is observed that the discomfort level in A is the highest, which also has the poorest indoor acceptability as seen in Figure 4. In general, occupants appear to be satisfied with their IAQ. However, occupants from building A seem to be dissatisfied with their air quality, which is evident from a score exceeding 5. Building B also registers a notably significant number of people dissatisfied with their indoor environment. Taking a scale of 1–5 as satisfactory indoor air acceptability, 6–10 as dissatisfied, it is seen from Figure 4 that a higher level of dissatisfaction exists among occupants for ‘past month’ condition than ‘here and now’. However, on the whole, people accepted the range of IAQ as reasonably good in all buildings except building A, as evident from Figure 5. While the percentage dissatisfaction of indoor air acceptability in A is about 50%, other buildings generally experience less than 20% dissatisfaction level. The level of overall acceptability for thermal comfort is also better than IAQ in general.

Performance indices

The various performance indicators such as the IPSI, ventilation index, energy index and the BSI are now

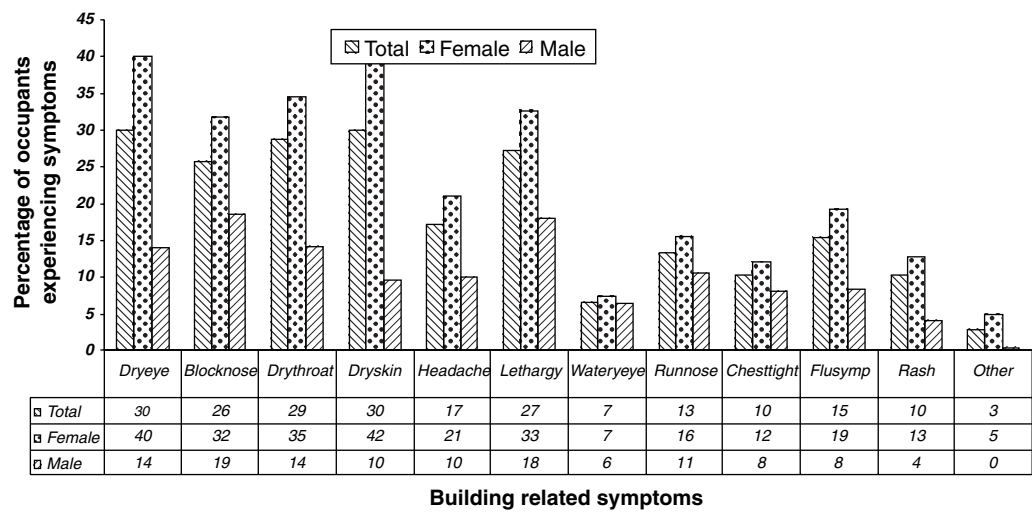


Fig. 2 A comparison of male and female building related symptoms (Mean of all five buildings) – past month (%)

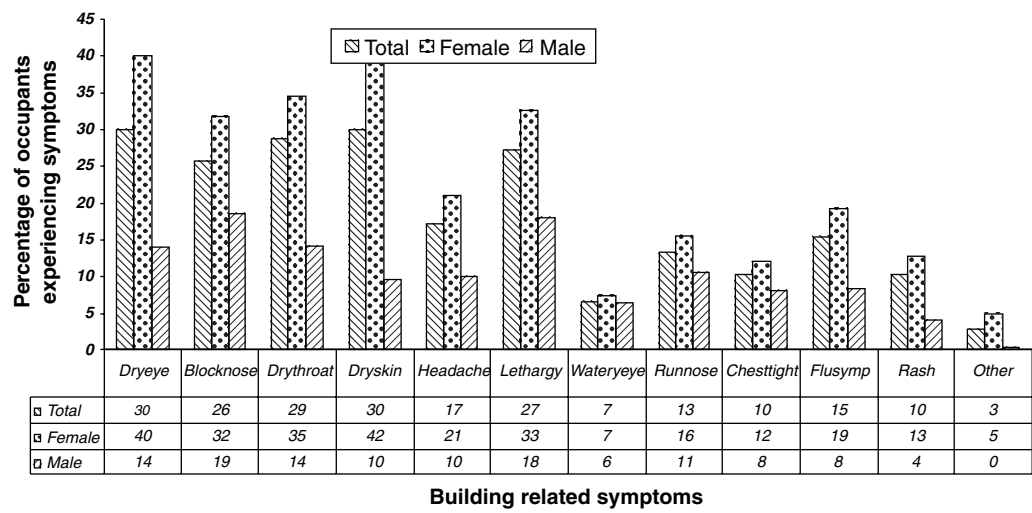


Fig. 3 A comparison of male and female building related symptoms (Mean of all five buildings) – here and now (%)

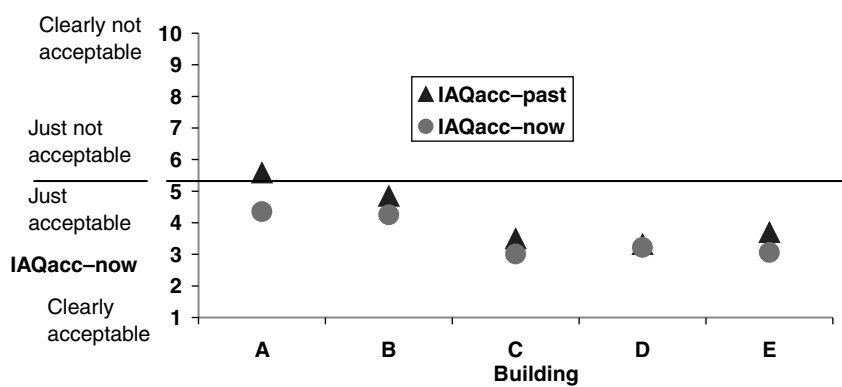


Fig. 4 Indoor air acceptability in buildings

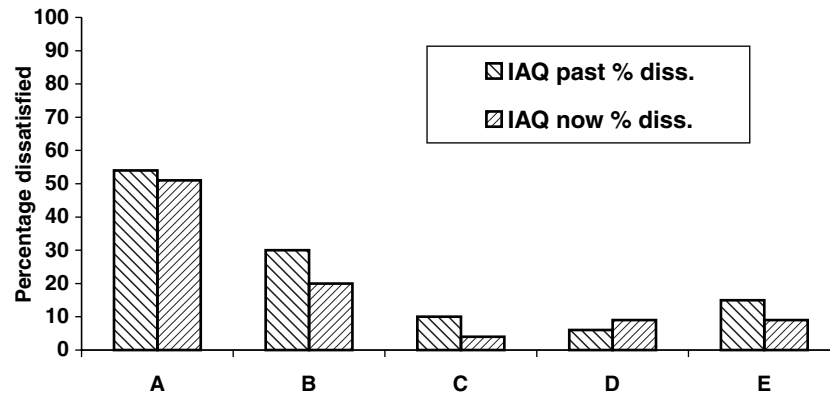


Fig. 5 Indoor air acceptability rated by occupants presented in % dissatisfied

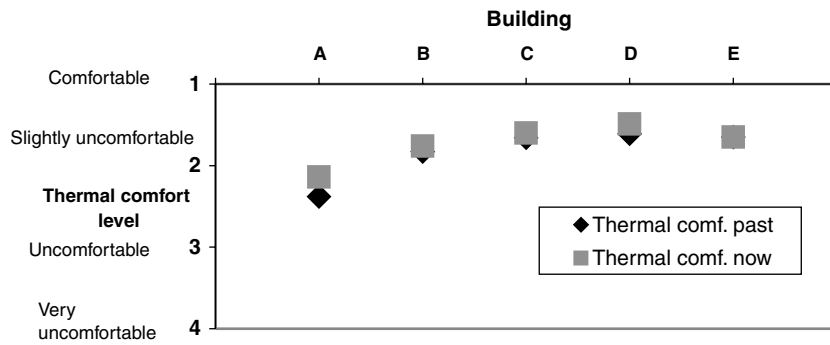


Fig. 6 Thermal comfort in buildings

presented and discussed. The former three indices are based on objective measurements while the latter is based on subjective measurements.

IPSI, ventilation index and energy index The IPSI, ventilation index and energy index developed for the five buildings are summarized in Table 5. An IPSI value of 2 corresponds to acceptability based on recommended guideline values for concentration levels of indoor pollutants. The uppermost value on the IPSI scale is 10, which corresponds to concentration levels of pollutants that are potentially harmful to human beings.

In building A, the IPSI values for three of the seven floors investigated are less than 2 while three of the remaining four floors have marginally higher IPSI values with one floor registering an IPSI value of 3.76. The IPSI in this building is attributed to either thermal comfort parameters or microbial contaminants. Based on a ventilation provision of 3.6 lps/person (0.36 lps/m^2) in the context of local guidelines, it is observed that the ventilation indices are generally adequate, ranging between 0.68 and 1.76 lps/m^2 . The energy index computed in this research project aims to address the chilled water energy consumption, which

Table 5 Summary of performance indices

Building	Zone	Indoor pollution standard index		Ventilation index (l/s m^2)		Energy index (kWh/m^2)	
		Zone	Mean	Zone	Mean	Zone	Mean
A	7 West	1.60	2.21	1.04	1.14	83.44	55.1
	6 West	1.42		1.66		61.24	
	6 East	1.86		0.83		–	
	5 East A	2.22		0.96		41.58	
	5 East B	2.13		0.68		41.58	
	4 West	3.76		1.06		38.80	
	3 East	2.45		1.76		63.86	
B	B13	1.34	1.80	0.52	0.60	20.78	20.8
	B36	2.24		0.52		–	
	B44	1.82		0.76		–	
C	C4A	1.58	1.47	0.51	0.69	14.29	16.0
	C4B	1.26		0.74		23.65	
	C11A	1.29		0.61		15.68	
	C11B	1.58		0.97		15.56	
	C17A	1.63		0.57		15.08	
	C17B	1.48		0.78		11.59	
D	D6	4.82	2.88	0.37	0.37	25.75	25.8
	D23	1.78		0.39		25.75	
	D30	2.03		0.37		25.75	
E	E18	4.72	3.12	0.09	0.28	7.96	12.1
	E24	1.96		0.40		20.22	
	E47	2.69		0.39		8.08	

could be seen as the main contributor to a certain level of thermal comfort and IAQ with a certain ventilation provision. It is to be noted that no absolute benchmark figures are available for such an energy index. Hence, the findings from this study are envisaged to provide indicative values for benchmarking energy indices and also provide some insight into the relative interaction between energy and IAQ issues in air-conditioned buildings. The energy indices for building A range between 38.8 and 83.4 kWh/m²/month. It can be seen from Table 5 that higher values of energy index are usually associated with higher values of ventilation index, which is expected. However, a strong correlation of high ventilation index with low IPSI, implying better IAQ, is not evident in all the set-ups of building A.

In the case of building B, the IPSI values are generally acceptable with reasonable ventilation index values. Although the energy measurements were available for only one of the three set-ups studied, it is seen from Table 5 that the energy index is relatively low, which may be attributed to efficient use of energy in providing an acceptable level of IAQ.

The IPSI values in building C are within the recommended limit of 2 and these values are based on either thermal comfort parameters or microbial contamination. The values for ventilation index suggest that reasonable amount of fresh air is provided to the various zones. It is also seen that the energy indices are significantly lower in comparison with building A, which would imply that acceptable IAQ is achieved with efficient use of energy.

In building D, two of the three set-ups investigated have acceptable IPSI values of 1.78 and 2.03, while one set-up has a high IPSI value of 4.82 (based on TVOC). The sources of TVOC are mainly indoor and it is noted that renovation activities may have been carried out immediately preceding the measurements. It is also observed from Table 5 that the ventilation index is low when compared with buildings A, B, and C, implying that dilution due to fresh air may be inadequate. A low energy index of 25.75 kWh/m²/month is observed in this building.

The IPSI values of building E are generally high and are attributable to elevated TVOC levels. It is evident from Table 5 that ventilation indices are quite low and this clearly indicates that fresh air provision is inadequate. The energy index values for this building are the lowest among all the five buildings and this is logical when analyzed in the context of reduced fresh air quantities. It would appear that increasing the ventilation provision is, perhaps, the most viable recommendation for building E, which would result in lower IPSI values and increased energy index values.

Building symptom index (BSI) The BSI values for the five buildings are indicated in Table 6. In analysing the data pertaining to symptoms presented in Figures 2

Table 6 BSI For five audited buildings in Singapore

	Building					Mean
	A	B	C	D	E	
Total						
BSI ff (full)	2.77	2.11	1.56	1.8	1.53	1.95
BSI fs (short)	1.76	1.45	1.1	1.38	1.07	1.35
BSI sf (full)	2.08	2.10	1.80	1.62	1.59	1.84
BSI ss (short)	1.49	1.5	1.23	1.42	1.22	1.37
Past month						
BSI ff (full) male	2.60	1.78	1.46	1.10	0.91	1.57
BSI ff (full) female	2.75	2.24	1.70	2.08	1.84	2.12
Here and now						
BSI sf (full) male	1.26	1.59	1.28	0.44	0.97	1.11
BSI sf (full) female	2.23	2.32	2.67	2.04	1.83	2.22

and 3 and the BSI, linear regression, polynomial, logarithmic, powered and exponential statistical relationships were carried out with most variables and linear correlation data significant at a 5% level were used for analysis.

There is a reasonable variation among the five buildings for the various BSI values. When the total BSI values for full and short lists during 'past month' and 'here and now' periods are considered, the largest BSI (typically building A) is about 1.3–1.8 times that of the least BSI (typically building E). This variation becomes significantly greater when the analysis is distinguished between male and female occupants. While the variation in female BSI values is still about 1.6 between the largest and the smallest BSI, it is about 3 in the case of male BSI values. It is observed that the mean BSI value for female occupants is about 26 and 50% greater than those for male occupants for 'past month' and 'here and now' periods, respectively. It is thus evident that female occupants exhibit more building-related symptoms than male occupants, a phenomenon also observed in several earlier studies (Bluyssen et al., 1996; Womble et al., 1994; Stenberg et al., 1994).

Correlation studies were attempted on the various symptoms, indoor air acceptability, thermal comfort and BSI. A correlation attempt of environmental acceptability (IAQ and Thermal Comfort) between 'past month' and 'here and now' periods is presented in Figure 7. An interesting observation is made from Figure 7, which reveals that the acceptability of IAQ for both 'past month' and 'here and now' periods is significantly governed by the occupants' perceived level of THERMAL COMFORT. The perception of IAQ between 'past month' and 'here and now' periods is about the same as can be seen from the highly significant correlation obtained in Figure 8.

The occupants' perception of the indoor air acceptability also appears to be significantly governed by their perceived level of stuffiness in the indoor environment, which is presented in Figure 9. It is, however,

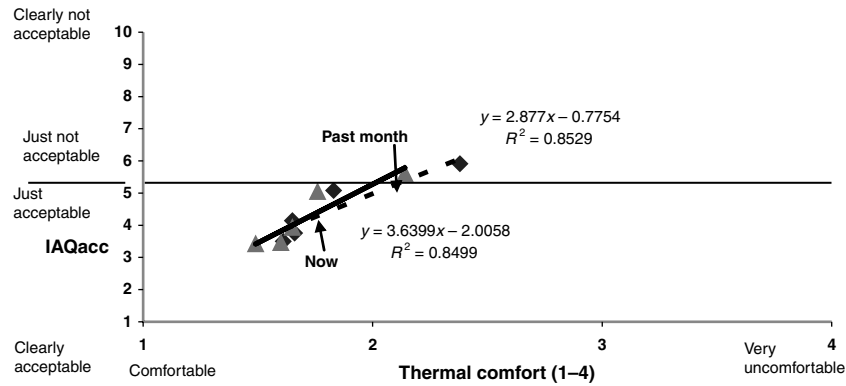


Fig. 7 Indoor air acceptability versus thermal comfort

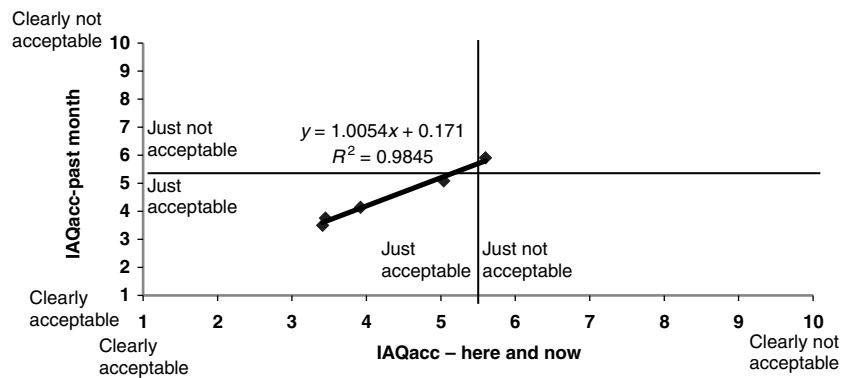


Fig. 8 Indoor air acceptability – past month versus indoor air acceptability – here and now

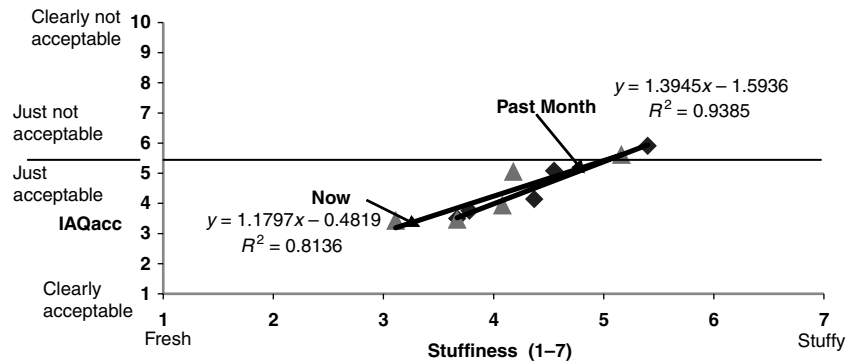


Fig. 9 Indoor air acceptability versus stuffiness

to be noted that the impact is greater for 'past month' in comparison with 'here and now'. Their perception of the indoor air acceptability is less significantly governed by their perception of indoor air odor, as seen in Figure 10.

The relationships among the four different types of BSI values are presented in Figures 11 and 12. There is a fairly significant relationship between BSIf and BSIfs as well as BSIfs and BSIs, presented in Figure 11. Based on full list of symptoms, the occupants' perception in the 'past month' is quite similar to that

experienced in 'here and now'. This relationship improves significantly when it is analyzed using the short list of symptoms. It can be reasonably concluded from Figure 11 that the occupants' perception of symptoms experienced are fairly similar both in the 'here and now' and 'past month' responses.

There is almost a strong relationship between BSIf and BSIfs, as evident from Figure 12. This suggests that a majority of the symptoms recorded during the 'past month' belong to the six most prevalent symptoms for the five buildings and that occupants'

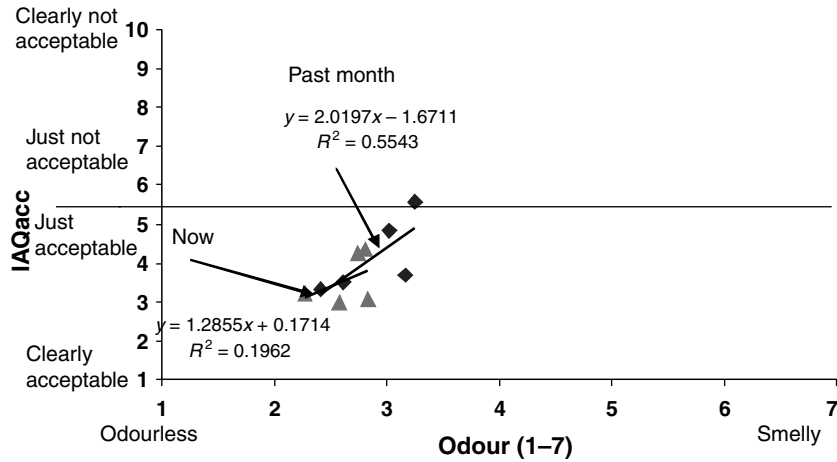


Fig. 10 Indoor air acceptability versus indoor air odour

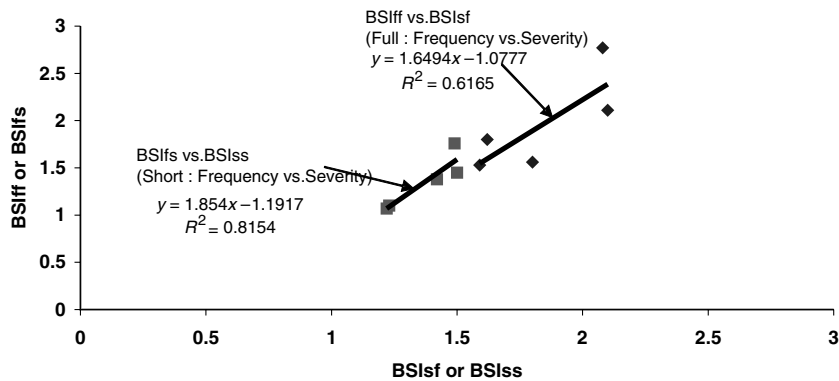


Fig. 11 BSIf vs BSIsf and BSIs vs BSIs

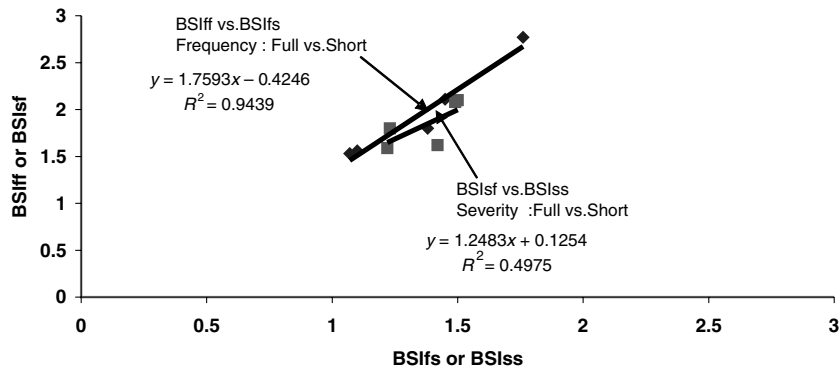


Fig. 12 BSIf vs BSIsf and BSIs vs BSIs

perception of the six most prevalent symptoms dominated the full list of symptoms. However, the same cannot be stated strongly for the symptoms experienced 'here and now', as evident from Figure 12. There is only a reasonable correlation between BSIsf and BSIs.

A strong correlation between BSI and THERMAL COMFORT, particularly for 'past month', is evident

from Figure 13. Similarly, BSI and IAQ are strongly correlated for both 'past month' and 'here and now' periods as evident from Figure 14. This suggests that the occupants' perception of building-related symptoms is significantly determined by their perception of either the thermal comfort or the IAQ.

Attempts to correlate between subjective data such as BSI and IAQ acceptability and objective data such

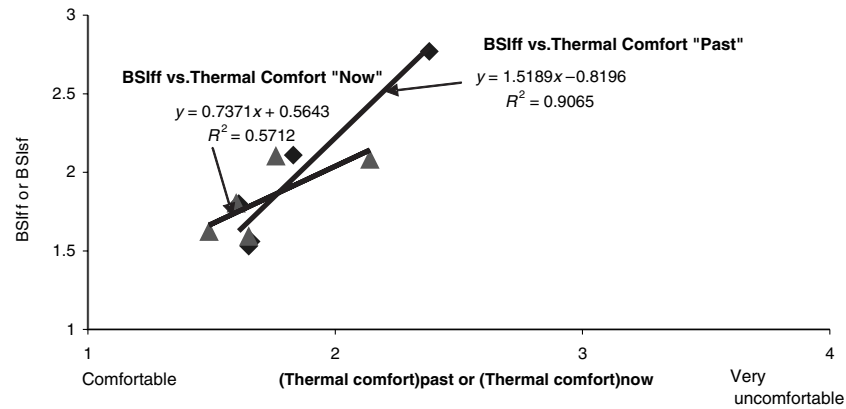


Fig. 13 BSI versus thermal comfort

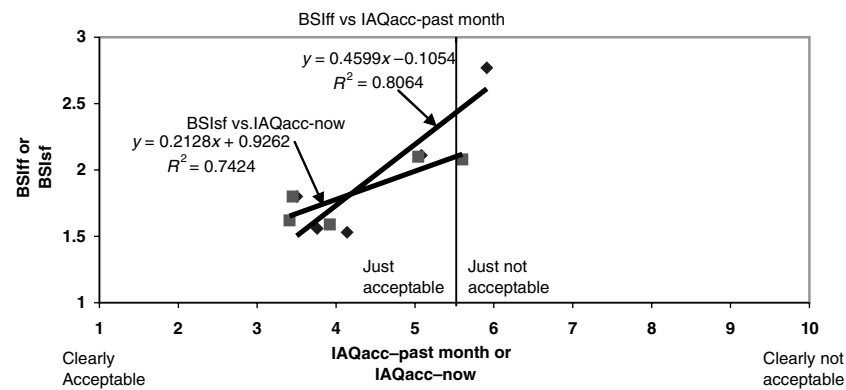


Fig. 14 BSI versus indoor air acceptability

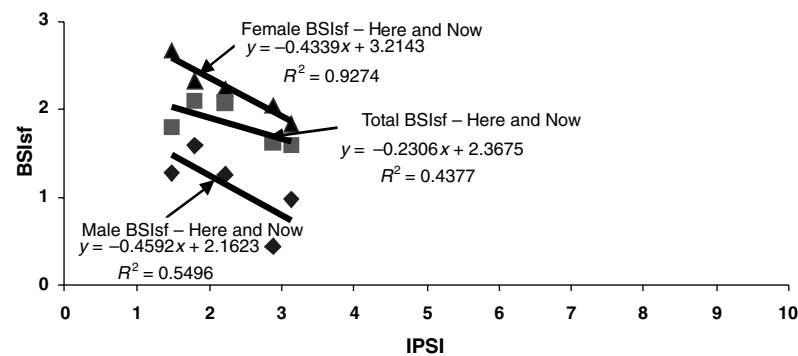


Fig. 15 BSIsf (here and now) versus IPSI

as IPSI, ventilation index and energy index were not very successful. A reasonable correlation coefficient between BSIsf and IPSI in Figure 15 suggests that the occupants' perception of their indoor environment and the symptoms experienced are not quite substantiated by objective measurements. In fact, when the IPSI is

lower, the occupants appear to be responding with more symptoms as evident from larger BSIsf values. While ventilation index is inversely proportional to IPSI, as would be expected, it is observed from Figure 16 that the correlation coefficient is not strong. IPSI does not appear to correlate well with indoor air

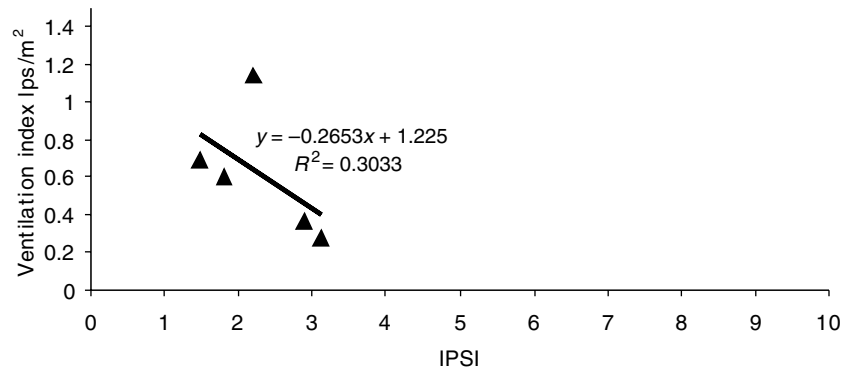


Fig. 16 Ventilation index versus IPSI

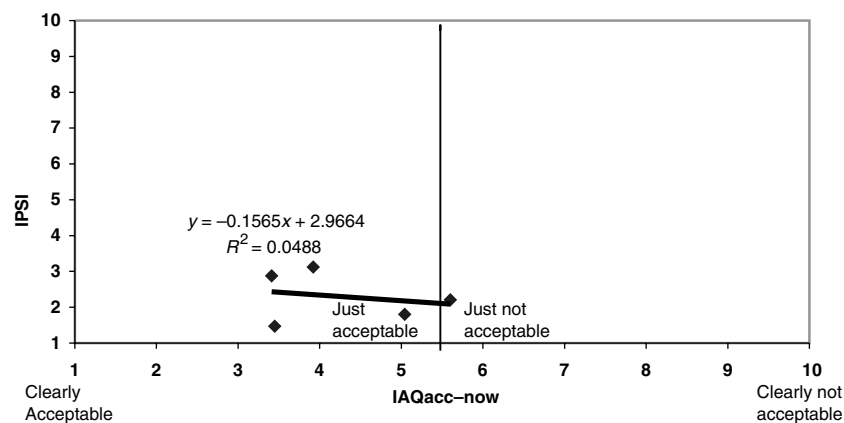


Fig. 17 IPSI versus indoor air acceptability – here and now

acceptability as evident from Figure 17, which again reiterates the fact that occupants' perception and measured environmental parameters are quite different. It is also observed that energy index and IPSI do not correlate in any significant manner.

Whilst it is reasonable to expect an association between IPSI and BSI, several confounding factors are believed to have an influence, resulting in a lack of clear correlation. First, the lack of detailed information on dose-response studies of indoor pollutants poses significant difficulty in establishing adequate number of breakpoints on an arbitrary IPSI scale of 0–10, which is the fundamental basis of the development of IPSI. Secondly, recommended TLVs in IAQ standards/guidelines emerge as a consensus, based on established industrial values from an occupational safety and health perspective. Although such indoor TLVs are generally lower than that for industrial settings, their direct relevance in non-industrial indoor environments, such as an office environment, is yet to be definitively established. Previous studies have shown that occupants exhibit SBS symptoms in sub-TLV exposures, resulting in BSI being a more

sensitive instrument than the proposed IPSI (Bluyssen et al., 1995; Sekhar et al., 2001). Thirdly, it is argued that occupants in an office environment are to be seen as responding to a 'cocktail effect' of various indoor pollutants, while an IPSI value, by its very definition, is pollutant dependent. It may then be argued that such inherent discrepancy between occupant responses and objective measurements of independent pollutant concentration levels is not surprising. The underpinning hypothesis postulated is that the occupants' perception of symptoms experienced as well as environmental acceptability is quite distinct from IAQ acceptability determined from empirical measurements of indoor pollutants, and that both IPSI and BSI are to be investigated to evaluate overall IAQ acceptability.

Conclusions

An elaborate collection of data, employing the integrated IAQ-energy audit methodology, carried out in Singapore is presented in this paper, in which an extensive database of concentration level of

various indoor air pollutants for typical office buildings employing conventional air-conditioning system designs with high recirculation rates has been obtained. Both objective and subjective measurements have been used. Using the local IAQ guidelines (ENV, 1996) whose TLVs are similar to those specified in the American and European standards, as a yardstick for comparison, the prevailing IAQ 'signatures' of buildings have been compiled and compared. Tracer gas techniques have been employed to determine the various ventilation characteristics such as the air change rates, fresh air provision and AEE values. In assessing the acceptability of IAQ, the subjective responses obtained through a detailed IAQ questionnaire have been analyzed to obtain occupants' perception of their indoor environment. The computation of chilled water energy consumption for different buildings and the corresponding IAQ signatures have provided an insight into the interaction between IAQ and energy issues in tropical buildings.

The computation of performance indices may be construed to be quick assessment tools in exploring the integrated issues of IAQ and energy consumption in buildings. IPSI is one of the performance indices developed in this research study. Other performance indicators used in the analysis are the energy index and the ventilation index. The ventilation provision is computed as a ventilation index in terms of lps/m² for the various buildings. The energy index computed in the study aims to address the chilled water energy consumption, which could be seen as the main contributor to a certain level of thermal comfort and IAQ with a certain ventilation provision.

A summary of the mean values of all the measured and computed parameters is presented in Table 7. The

figures presented in Table 7 are average values for the audited buildings and although large differences between individual buildings were observed, it is to be noted that they generally meet the requirements of the local guidelines in Singapore. In the case of thermal comfort parameters, it may be noted that despite the mean values of operative temperatures for each building being reasonable, certain measured locations indicate air temperatures in the lower range of the recommended values. A slight increase in these temperatures would imply an energy saving potential. The monthly energy consumption per air-conditioned floor area varies by a factor of 4.6 for the least energy consuming building to the most energy consuming building, which shows a reasonable theoretical economy potential as well as a significant diversity of conditions. The air temperatures measured in the occupied spaces of the audited buildings were generally in the lower range of the recommended values, and raising the air temperature set-points could provide an energy saving potential. Among the five buildings studied in this project, four of the buildings did not exhibit any clear relationship of poor IAQ and low energy consumption or vice versa. However, one building did indicate high IPSI values attributable to elevated TVOC levels, which was also associated with low ventilation provisions. The energy index of this building was also one of the lowest among all the buildings.

In the case of subjective data analysis, occupants generally appear to be satisfied with their IAQ. The level of overall acceptability for thermal comfort was better than IAQ in general. Female occupants register significantly higher building-related symptoms in comparison with their male counterparts in both the 'past month' and the 'here and now' periods. The BSI values differ quite significantly among the five buildings studied and the mean BSI values for female occupants are up to 50% greater than male BSI values, implying a greater prevalence of building-related symptoms among female occupants. While BSI values are found to correlate among them as well as with IAQ and THERMAL COMFORT acceptability, no significant correlation was observed between BSI and IPSI. This would suggest that the occupants' perception of symptoms experienced as well as environmental acceptability is quite distinct from IAQ acceptability determined from empirical measurements of indoor pollutants, which reinforces the complex nature of IAQ issues.

Acknowledgements

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Table 7 Summary of measured and computed parameters

	Singapore audit (range of mean values of five buildings)
Carbon dioxide	506–797 p.p.m.
Carbon monoxide	0.8–1.6 p.p.m.
Total volatile organic compounds (TVOC) (toluene equivalent)	0.35–2.25 p.p.m.
Particulate matter	20–35 µg/m ³
Operative temperature	22.3–24.2°C
Air velocity	0.13–0.18 m/s
Outdoor air change rate	0.34–2.6/h
Outdoor air flow rate	0.37–1.14 lps/m ²
Global air exchange effectiveness (AEE _G)	0.96–1.8
Occupant level air exchange effectiveness (AEE _{OL})	0.74–1.11
Localized air exchange effectiveness (AEE _L)	0.58–1.23
Indoor pollutant standard index (IPSI)	1.47–3.12
Energy index	12.1–55.1 kW h/m ² /month

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